

Project Proposal: SafePath Analytics

Empowering Public Schools with Independent Digital Safety Audits

The Problem: Information Asymmetry in Education

A severe structural gap exists between technology platforms and public education. Major tech companies utilize advanced algorithms and vast datasets to optimize youth engagement. Conversely, local school districts and educational policymakers operate in a technical vacuum. When addressing the escalating youth mental health and attention crisis, schools lack the independent tools necessary to measure exactly how proprietary algorithms are impacting their specific student populations.

The Solution: SafePath Analytics

SafePath Analytics is an open-source analytical framework built exclusively for public school systems. It acts as an independent "audit layer," translating complex data science into actionable insights for educators, without compromising student privacy.

1. The Data Inputs (What We Analyze)

To ensure strict compliance with the Family Educational Rights and Privacy Act (FERPA), the system operates entirely on anonymized and aggregated datasets. The framework ingests two primary data streams:

- **De-identified Digital Footprints:** Aggregated, anonymized device usage statistics (e.g., screen time volume, application category categorization, platform engagement frequency) sourced from school-managed networks or opt-in district devices.
- **District-Level Baseline Metrics:** Anonymized, high-level school data regarding attendance rates, aggregate disciplinary reports, academic performance baselines, and counseling resource utilization.

2. The Analytical Engine (How We Analyze It)

Unlike basic survey tools that only show correlations, SafePath utilizes advanced data science methodologies – specifically **Causal Inference** models (such as Propensity Score Matching and Difference-in-Differences). This allows the tool to filter out confounding

variables and isolate the *causal* impact of algorithmic platform exposure on student well-being, effectively replicating rigorous tech-industry testing within a public sector environment.

3. Actionable Insights (What Schools Get)

The final output is not raw data, but highly visual, vendor-neutral dashboards and reports designed for school administrators, IT directors, and counselors. Key deliverables include:

- **Policy Dashboards:** Clear visualizations demonstrating how specific types of digital engagement affect focus metrics or mental health resource utilization across different grade levels.
- **Infrastructure Recommendations:** Data-backed guidelines empowering district IT administrators to optimize firewall configurations and app restrictions on school-issued devices (e.g., Chromebooks, tablets).
- **Targeted Resource Allocation:** Identifying specific digital risk factors so educators can deploy timely interventions, mental health resources, or tailored digital literacy curricula where they are needed most.

Current Status & Roadmap

SafePath Analytics has advanced from conceptual design to tangible execution. The architectural framework for a Zero-Trust Data Transfer pipeline is complete, and the foundational open-source codebase is actively being developed. Furthermore, independent public health researchers have rigorously validated the mathematical framework (Local Differential Privacy) and causal inference models, confirming their operational viability for public sector deployment. As we transition to the deployment phase, **we are actively seeking preliminary discussions with forward-thinking school districts, policymakers, and civic leaders to align our ongoing development with localized educational needs.**